

Data Analysis Questions

Health Insurance Portfolio Dataset — Question Bank by Analysis Type

A categorized set of questions this dataset can answer, from simple descriptive stats through to the predictive models. Useful as a portfolio/interview scope reference, or as a checklist while doing EDA — pick one or two questions per category rather than trying to answer everything.

1. Descriptive — what's happening in the book of business

Surface-level portfolio stats. Good starting point before anything else.

- How is portfolio size trending across the 3 periods — growing, shrinking, or flat? (compare *new_business* counts vs. lapse counts per period)
 - What's the mix of individual vs. collective policies (*type_policy*), and has that mix shifted year over year?
 - Which *type_product* (Dental / Premium / Standard / International) is most common, and is that changing over time?
 - Which *distribution_channel* brings in the most business?
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2. Diagnostic — why lapse/churn is happening

Explains the drivers behind the lapse target before you model it.

- What actually drives lapse — age, product type, channel, seniority? (group lapse rate by each feature)
 - Do brand-new customers (*new_business* = Yes) lapse more in their first year than renewing customers?
 - Is there a "seniority cliff" — a specific tenure point where lapse risk jumps or drops sharply?
 - For collective policies, does one family member lapsing tend to happen alongside others on the same *ID_policy*? A real operational insight most people don't check.
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3. Diagnostic — why claims cost/risk is happening

The actuarial side: frequency vs. severity, and whether pricing tracks risk.

- What's the loss ratio (*cost_claims_year* / *premium*) overall, and by segment — is any product or channel actually losing money?
 - Which segments have the highest claims *frequency* (*n_medical_services*) vs. highest claims *severity* (cost per claim)? These are different problems.
 - Does *premium* pricing actually track risk — are there segments where people are underpriced relative to their claims cost?
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4. Geographic & socioeconomic

Uses the postal-code / municipality concentration and income/education indices.

- Does market penetration (*IICIMUN*, *IICIPROV*) correlate with lapse rate — are you "stickier" in markets where you're already dominant?
 - Do higher-income postal codes (*C_GI*, *C_II*) show different utilization or claims patterns than lower-income ones?
 - Does municipality size (*C_H*: H1 rural → H6 large city) affect claims cost — e.g. rural areas having less access and different utilization?
 - Is there a regional/climate pattern (*C_C*) in claims cost or product mix?
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5. Time / cohort analysis

Follows the same insured across periods using the ID field.

- Track a cohort that joined in period 1 — what fraction are still active by period 3? (a simple retention curve)
 - Does *seniority_policy* (contract age) protect against lapse better than *seniority_insured* (individual tenure)?
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6. Predictive — where the two models fit in

Everything above becomes the feature list feeding these.

- "Will this insured lapse, and which kind?" — the *lapse* classifier (1: early, 2: active, 3: expiry).
 - "What will this insured cost in claims next year?" — the *cost_claims_year* regressor.
 - A diagnostic finding (e.g. "collective policies lapse together") becomes an engineered feature for these models, not just a standalone factoid.
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7. Data-quality questions

Worth asking explicitly before trusting any of the above.

- Any rows where *date_lapse_insured* is filled but *lapse* == 2 (active)? A data inconsistency worth flagging.
 - What fraction of rows have *exposure_time* < 1, and does that cluster around policy start/end dates as expected?
 - Do the geographic/socioeconomic columns go missing on the same rows (one root cause) or independently (several)?
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Tip: for a portfolio writeup or interview, pick one question per category, answer it with an actual chart/number from the data, and add one sentence of "so what" — that shows analytical range beyond just training a model.